

ANAHAT TRANSFORMATIONS

Gopal Park, Behind Image Sonography Clinic,
Paud Phata Rd, Erandwane, Pune 411004

ABC of Music

We've explored both scientific and traditional perspectives on sound. Music, however, transcends mere sound, becoming a source of pleasure and emotional resonance. Musical experiences are uniquely memorable, often linked to specific events, people, or periods. This association can trigger vivid recollections, both positive and negative, evoking a complex mix of joy and melancholy. While time erodes the original context, music's power to evoke these memories remains.

Music therapists face the challenge of selecting pieces that avoid exacerbating negative memories. Conversely, "goodtime music" can be a powerful tool for rejuvenation and recovery, particularly for individuals experiencing anxiety or depression. The therapeutic use of melancholy, as noted by the English poet Shelley's observation that "the sweetest songs are those that tell of saddest thought," is a recognized approach to emotional healing.

Assuming unimpaired hearing, anyone can engage with music. Active listening allows one to discern the emotions conveyed, which are often complex and layered. Although music and noise originate from the same acoustic foundation, music is distinguished by its organized and pleasing nature. Unlike the erratic and unpleasant character of noise, music is structured with mathematical precision (rhythms and beats) and aesthetic refinement (tones and resonance). This ordered arrangement of tones, with defined pitch and patterns, enables the brain to process it as a meaningful and beautiful experience.

The Content of Musical Sound

When we concentrate on the makeup of musical sound, we begin to understand how it captivates us. Music uses aesthetically pleasing pitches and dynamic notes that move through scales, drawing our attention along its path. In parallel, rhythms trigger involuntary physical reactions, like nodding our heads or moving our bodies. The following paragraphs will break down the components of this immersive experience.

Musical Pitch

Pitch renders an "identity", unique to a tone. As we call a note high-pitched, it would really mean that we are exposed to more sound waves per second, as compared to the one we may call "low-pitched". Pitch, which refers to the location of a sound in a tonal scale, gets classified into high or low, depending upon the speed of vibrations from the sound source.

It is interesting to note that the pitches used today have evolved through observations over several centuries. The present day standard of the "concert pitch" as agreed upon in international conferences held in 1939 and 1960 rules that the "A" directly above the Middle "C" has 440 (double) vibrations per second (440 Hz), which makes the Middle "C" 261.6 Hz.

It is interesting to observe that the frequencies of the Middle "C", located in the middle of the keyboard dwell more or less in the centre of the frequency of human conversation, which is anywhere between 200-400 Hz. However, the vocal pitch is never constant. It is invariably associated with one's voice constitution, which varies from male to female and also from person to person. At times due to fatigue or depression, the same voice could turn out to be much lower and the pitch

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range very much narrower than what is usual. On the other hand, when elated, the speaker's general tone of voice rises significantly. Haven't we come across the joyful interaction of the kids and teenagers, when holidays are announced from their routine studies or in a picnic mood?

As regards the norms of musicology, the Western system appears to insist on a much-respected (rather, mechanical) precision in respect of pitch in a classical musical composition. The Indian system is however, singer-friendly in the sense that there is a generous allowance for determining the pitch taking into consideration the natural endowment of one's voice.

High-pitched Notes

High-pitched notes correspond to a greater number of sound waves produced per second. Consequently, the frequency of oscillations, measured in cycles per second (Hz), is higher for high-pitched notes compared to low-pitched ones. This increased frequency results in sound waves being compressed, leading to shorter wavelengths.

Jacques de Liège, a 14th-century musician and musicologist, observed that vocal pitch increases in intensity as it rises. He likened this intensification to a painter's application of color or white to a canvas. He also noted that singers expend more breath when producing high notes, which are inherently stronger and more brilliant, than when producing low notes.

Frequencies exceeding 20,000 Hz, beyond human hearing, are utilized by bats for echolocation, enabling them to navigate and hunt in darkness. Researchers at London's Queen Mary College have recently recorded these ultrasonic frequencies.

Low-pitched Notes

Low-frequency sounds propagate over greater distances than high-frequency sounds. Historically, drums were used for public announcements due to this characteristic. In forts like Golconda, "bheris" (drums made of animal skin and wood, capable of producing low-pitched notes) facilitated long-distance communication through their low-pitched tones.

European church organs can produce frequencies as low as 10 Hz. While often inaudible, these frequencies can be felt as vibrations and cause physical resonance in building structures, including stained glass windows (vitrines).

Natural examples of low frequencies include thunder (20-40 Hz) and blue whale vocalizations (16 Hz), contrasting with the high-frequency squeaks of mice (3000 Hz).

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Tone

A tone, within the realm of music, is a sound defined by its steady vibration, a constant pitch that invites harmonious connection with others. Tones are living entities, shifting and evolving with variations in frequency, duration, strength, and harmonic complexity. Honoré de Balzac saw them as colors, each painting its own emotional hue.

Ancient Indian musical thought classified tones by their origin, revealing a deep understanding of their nature: the human voice (*sariraja*), the plucked strings (*nakhaja*), the breath of wind (*vayuja*), the rhythmic beat (*charmaja*), the shimmer of metal (*lohaja*), and the drawn bow (*dhanvija*).

Roger Scruton, in "Aesthetics of Music," suggests that a tone's meaning is found in its context, a response to what came before and a suggestion of what is to follow, creating a continuous musical narrative.

The sounds we experience are rarely pure, but rather a rich tapestry of interwoven tones and overtones.

Timbre or Tonal Quality

We use the French word "timbre" to describe the unique color or quality of a tone, a characteristic our ears are particularly attuned to. It's this timbre that enables us to distinguish the richness of a human voice from the sound of a musical instrument, and to discern the difference between male and female voices.

The form of sound wave vibrations dictates timbre. The richness and complexity of a tone's quality are shaped by the quantity and prominence of the harmonics it contains.

Octave Equivalence: The Relationship between Pitches

Musicians recognize that pitches with different fundamental frequencies can sound remarkably similar. For example, a 440 Hz pitch and an 880 Hz pitch share a perceived similarity. This phenomenon, known as "octave equivalence," occurs when pitches are related by powers of 2.

In musicology, octave-equivalent pitches are grouped into the same pitch class. This allows for the analysis of relationships between notes based on their pitch. Since notes can be categorized as higher, lower, or equal in pitch, the concept of an interval—the distance between two pitches—becomes relevant. Intervals are defined by frequency ratios.

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Musical Scale

A musical scale is essentially a collection of pitches arranged around a central starting note, called the tonic. When a tonic is chosen, a scale can be built by defining the intervals between the tonic and the other pitches.

Generally, scales deal with pitch classes, meaning they repeat when you reach an octave higher or lower than the tonic. This allows us to define a scale within a limited range of one octave.

Scales use a select number of pitches, rather than all possible ones. The notes chosen often have simple frequency ratios with the tonic, which helps create the sense of consonance (harmony) and dissonance (tension). In essence, everything is relational to the tonic.

In Indian music, while the specific frequency of the tonic is not fixed, the scale precisely locates the other notes once the tonic is established. The tonic's presence is often emphasized using a drone instrument such as the tanpura or shruti-box.

The Perception of Consonance and Dissonance

The terms consonance and dissonance help us understand whether certain combinations of musical notes sound harmonious. Consonant pitch combinations are generally more pleasing to the ear, and even infants and untrained listeners tend to prefer them. This preference for consonance is a guiding principle in the creation of many nursery rhymes.

Interval: The Distance between Pitches

The consistent pattern of intervals between notes defines the structure of musical scales. A universal method exists for calculating the distance, or interval, between pitches in any musical system. This interval is quantified as the ratio of the higher frequency to the lower frequency. For example, to find the interval between the second and third harmonics, you divide the third harmonic's frequency ($3f$) by the second harmonic's frequency ($2f$), yielding a ratio of $3/2$, or 1.5. This means the third harmonic's frequency is 1.5 times that of the second.

The German occultist Rudolf Steiner believed that the intervals of the Western musical scale held deeper, esoteric meanings.

Melody

Melody is the artful arrangement of tones, each flowing seamlessly into the next. Visualize it as a delicate necklace, strung with pearls of different sizes—each pearl a note. This musical form is shaped by the interplay of repeated notes or phrases, contrasting elements, and subtle variations. In its essence, a melody is a coherent series of notes that resonate harmoniously together.

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Rhythm

Rhythm, often considered the "heartbeat" of music, dictates the temporal flow of a melody, satisfying our innate desire for a consistent beat. Unlike pitch, rhythm deals with the time aspect of music.

Rhythm involves a range of elements, including accentuation, individual beats, the organization of notes into beats, beats into measures, and measures into phrases. These aspects are meticulously arranged in music, even in forms with "free rhythm," such as plainsong or simple bhajans.

Since ancient times, shamans have understood the therapeutic effects of tones and rhythms. Slow, repetitive rhythms, as found in lullabies, induce calmness, while faster, more varied rhythms stimulate activity.

Recent studies by neurologists and psychologists indicate that musical beats around 60 bpm can positively influence bodily rhythms, such as heart rate, breathing, and blood circulation. It's interesting to note that many traditional and ethnic songs across the globe are based on these rhythmic patterns.

Slow, iterative melodies, such as those in Gregorian chants, bhajans, and kirtans, have been shown to promote relaxation and cardiovascular well-being, contrasting with fast-paced, progressive music, which tends to energize the listener.

Harmony

We've established that melody is a series of pleasing tones, arranged sequentially. We also used the analogy of a single-strand pearl necklace to illustrate this concept.

This "single-strand" idea corresponds to "monophony" in music, a common feature in world music, particularly in traditional forms. Western music expanded this concept by developing "polyphony," a "multi-strand" version, which has been popular since the 13th century.

Polyphony involves the simultaneous sounding of multiple notes, producing a "vertical" musical texture distinct from a single melodic line. This is often called "harmony," where multiple melodic patterns surround a primary melody. By skillfully combining tones, rhythms, and harmonies, one can create a powerful resonance capable of addressing imbalances.

Joanne Crandall observes, "No natural law, based on acoustics or physics or the planet could determine harmony; it's our minds, conditioned by our cultural and personal preferences, could make that decision."

Counterpoint

Counterpoint is a musical innovation that significantly enriched our listening experience. This technique, mastered by European composers during the Renaissance and prominent in Baroque music, derives from the Latin phrase "punctum contra punctum," meaning "note against note." Counterpoint refers to the interplay between two or more distinct melodic lines, each with its own contour and rhythm, yet harmonically related. The adjective form, "contrapuntal," directly relates to the Latin origin.

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Agogic

The term "agogic," originating from the Greek "agoge," meaning melody, refers to variations in accentuation based on the musical phrase, rather than the regular beat. In a wider sense, "agogic" encompasses all expressive aspects of music, including changes in tempo, accents, pauses, and rhythmic flexibility.

Music, while often perceived as simple, reveals its complexity upon deeper inquiry. Although we appreciate it, music defies easy comprehension, presenting inherent contradictions: it's both localized and transcends location, focused and diffused, intense and expansive, specific and universal, stable and dynamic. These contradictions create a fertile ground for philosophical interpretation.

Hegel argued that music, unlike other art forms, lacks spatial existence and therefore, "objectivity." Frank Zappa compared musical performance to sculpting the air. Claude Levi-Strauss described music as an intelligible yet untranslatable language, highlighting the listener's experience of internal rhythm, a concept explored in Nada Yoga. Roger Scruton proposed that music's acousmatic domain is separated from the physical realm by a metaphysical barrier, with its own phenomenal space, time, and causality.

Music is essentially a reflection of our personal experiences, thoughts, and wisdom. It provides immediate emotional connection, bypassing direct explanation. Its dynamic nature mirrors life's fluctuations, striving to bring balance and spiritual elevation.

Music's inherent value lies in its continuous ability to provide energy and strength.

The Pattern

A musical pattern, in its most general definition, is a "structure"—be it melodic, rhythmic, or harmonic—that possesses notable prominence within a musical composition. It's essentially a mental blueprint or an "architectural vision." For example, a melodic pattern refers to a phrase or fragment that is reiterated, with or without alterations, throughout a musical work.

The traditional concept of musical patterns has led to extensive research in computational modeling and simulation in recent years. One research direction involves computational music analysis, which incorporates the concepts of patterns, repetitions, and similarities. Another area focuses on creating music-generating algorithms and systems that utilize predefined or learned patterns as foundational elements in the generative process.

The Intelligence

Musical intelligence allows individuals to understand and appreciate the nuanced meanings embedded in various musical components. These include pitch, individual notes, amplitude, texture, tone color, rhythm, melodies, metric patterns, harmony, structural patterns, and expressive variations. It encompasses the comprehension of their intricate arrangements, combinations, and the strategic use of silence. Furthermore, it empowers individuals to create organized, pitched sequences as a form of self-expression or interpersonal communication.

The act of singing notes with precise pitch necessitates mental focus and intellectual involvement. Consequently, music provides a training ground for developing cognitive abilities.

Ancient Schools of Music

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Archaeological excavations of Paleolithic human bones across the globe have uncovered specimens with precisely drilled lateral holes, indicating their use as flutes. These resemble the Japanese shakuhachi and suggest musical instrument use by early humans.

The earliest recorded musical expressions are found in the 3,000-year-old Indian Sama Veda and the equally ancient cuneiform texts from Ur. Archaeological discoveries from the Indus Valley civilization (3000 B.C.) include seven-holed flutes and a variety of stringed instruments.

Indian music, one of the world's oldest musical traditions, meticulously documented its grammar and codes in ancient texts. In ancient Greece, mixed-gender choruses performed during spiritual ceremonies and for entertainment. Greek musicians and singers played a vital role in theatrical productions, and music was integrated into children's education.

Al-Farabi (872-950) authored the significant Arabic text, "Book of Music" (Kitab al-Musiqa), and invented numerous musical instruments. Arabic music continues to adhere to the pure Arabian tonal system he developed.

Medieval and Renaissance Periods

Numerous literary and archaeological finds indicate that music was a prevalent art form in human settlements throughout the world. The ancient Shiva temple in Thanjavur showcases silent sculptures depicting dancers and musicians performing with their instruments.

Europe experienced the rise of several schools of polyphony. In contrast to the predominantly devotional nature of Indian music (Bhakti sangeet), much of the surviving 14th-century European music was secular. However, by the mid-15th century, European composers began to incorporate smooth polyphony into both sacred musical compositions (mass, motet, and laude) and secular forms (chanson and madrigal).

It is widely acknowledged that the invention of the printing press played a crucial role in accelerating the spread of these musical styles.

Baroque Period

The beginning of opera around 1600, along with the progression of counterpoint in music, signaled the end of the Renaissance and the start of the Baroque period. This musical era ended around 1750, marked by the passing of influential composers George Frideric Handel and Johann Sebastian Bach. Baroque composers, and particularly Bach, developed the German musical style through strong contrapuntal techniques and a very detailed control of harmonic and motivic organization. Bach's contributions within music are often compared to the significance of Shakespeare's contributions to English literature, and Newton's within Physics.

During the Baroque period, important music styles were created, and developed later on into expanded musical forms, these include: the Fugue, the Invention, the Sonata, and the Concerto.

Simultaneously in India, this period saw the peak of compositions by the South Indian Carnatic musical trinity of Thyagaraja, Dikshitar, and Syama Sastry. These three South Indian composers lived during the same time period as the Western classical composers of Bach, Beethoven, and Mozart.

Classical Period

The Classical period in music is defined by its homophonic style, where a clear melody is supported by accompanying harmonies. Melodies from this era were designed to be similar to vocal lines,

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making them easily singable. Instrumental music experienced significant growth, building upon the forms established in the Baroque period, such as the sonata and concerto, while also introducing the symphony. Central composers of this period include Franz Joseph Haydn and Wolfgang Amadeus Mozart. Interestingly, almost two centuries after his death, Mozart's music is now being acknowledged for its therapeutic qualities.

Romantic Period

Ludwig van Beethoven and Franz Schubert acted as transitional figures, leading into the Romantic period by expanding existing musical genres, forms, and functions. During the Romantic era, the emotional and expressive qualities of music became paramount, overshadowing the emphasis on technical skill and adherence to tradition. The late 19th century witnessed a significant increase in orchestra size and the integration of concerts into the lifestyle of the urban elite. Later Romantic composers produced more intricate and often substantially longer musical compositions. For instance, the integration of counterpoint with complex harmonic structures, the expanded use of dissonance, and the creation of dramatic tension and resolution elevated the musical experience to new levels.

The 20th Century music experience

The 20th century brought significant changes to music consumption through the widespread adoption of radio and the development of recording and distribution technologies. Art music in this period was characterized by experimentation, with key figures such as Claude Debussy, Igor Stravinsky, Arnold Schoenberg, and John Cage shaping its direction.

In India, the arrival of cinema and radio led to the popularity of film music ("filmy geet"), which addressed social issues and blended traditional Indian melodies with Western musical elements. Some Bollywood compositions drew inspiration from Western classical music and Indian folk traditions.

The 20th century also witnessed the development of Jazz in the West, which gained a substantial following, and the subsequent global phenomenon of rock music.

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Contemporary Schools

Contemporary music, in both Indian and Western contexts, is a product of modern urban life, reflecting anxieties related to time scarcity, financial pressure, social anonymity, intense competition, globalization, and workforce mobility. These factors have contributed to a musical landscape often characterized by superficiality, mechanical sounds, and a focus on themes of sexuality, passion, and, at times, violence, with a perceived disregard for traditional human values.

This trend toward rhythm-driven music and high-decibel levels suggests a shift away from melody and emotional depth, prioritizing action over reflection. This can be interpreted as a cultural transition from social connection to self-centeredness, with defiant music mirroring social and familial fragmentation. The emphasis on material gain and individual success has arguably marginalized traditional values like love and compassion.

The music reflects a social environment where financial success is often prioritized over empathy, leading to a culture of superficial interactions. The aggressive rhythms, amplified volume, and often incoherent lyrics symbolize the instability and uncertainty inherent in contemporary life.

Music therapists debate the relative therapeutic merits of traditional music (folk, tribal, classical) and contemporary forms. While traditional music offers structured, potentially calming patterns, contemporary music may provide a necessary cathartic outlet for the suppressed emotions of urban dwellers. The choice between prioritizing material wealth and personal well-being remains a significant consideration.

It is important to acknowledge that all musical forms, consciously or subconsciously, are linked to human emotions. Therefore, a broad and inclusive approach to music is essential, as all genres can potentially offer therapeutic benefits at various stages of life.